V.V.P. ENGINEERING COLLEGE
DEPARTMENT OF COMPUTER ENGINEERING
List of Practical - A.Y. -2025-2026

Staff: Miss Brinda Chanv, Mr. Bhavesh Chhatrala, Miss Gunjani Vaghela

Subject: Object Oriented Programming (BE04000231)

Lab No.	Name of Experiment	CO	CL
1	**Program List 1:** 1) Write a program to display first n prime numbers. 2) Write a program that takes the lengths of three sides of a triangle as input. Calculate and print the area of the triangle. Ensure that the program validates if the given side lengths can actually form a triangle (sum of any two sides must be greater than the third side) before calculating the area (use Heron's formula: Area = sqrt(s*(s-a) *(s-b) *(s-c)) where s = (a+b+c)/2). 3) Write a Java program that prompts the user to enter a single letter (character). Determine whether the entered character is a vowel (a, e, i, o, u, case-insensitive) or a consonant, and display the result. 4) Write a Java program that accepts a five-digit integer from the keyboard. Your program should then create a new number by adding one to each digit of the input number. For example, if the input is 12391, the output should be 23402 (note: 9+1=10, so it becomes 0 with a carry).	1	Ap
2	**Program List 2:** 1) Write a program to validate email address. 2) Write a program to validate GTU enrollment no. 3) Write a program to check given Strings are anagram or not.	2	Ap

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3	**Program List 3:** 1) Define a class Student with name of student, department, and 3 subject marks as data members. Define methods to enter details of 5 students, print details of the entered students and sort details of the students on the base of their percentage. 2) Create an Employee class with private instance variables for employeeName (String) and employeeSalary (double). Implement public methods readEmployeeData() (to take input from the user) and displayEmployeeData() (to print the employee's name and salary). Demonstrate object creation and method invocation in a main method. 3) Design a class Time with hours (int) and minutes (int) as data members. Include method setTime(int h, int m) to initialize the time and displayTime() to display the time. Implement a method addTime(Time t1, Time t2) that takes two Time objects as arguments, adds their hours and minutes, and stores the result in the calling object. Do not use constructors for initialization in this specific practical.	2	Ap

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4	**Program List 4:** 1) Create a Java class named University with a static data member totalStudents to keep track of the number of student objects created. Implement a static method getTotalStudents(). Also, include a static block to initialize a static variable (e.g., universityName) and an instance block to print a message when an object is created. Demonstrate their execution order. 2) Write a Java program that demonstrates method overloading to calculate the volume of different 3D shapes. Implement overloaded methods named calculateVolume() for a Cube (takes one side length), a RectangularCube (takes length, width, height), and a Sphere (takes radius). 3) Write a Java program to demonstrate method overriding in an online payment system. Create a superclass Payment with a method processPayment(int amount). In the subclass CreditCardPayment, override the method to print "Payment of Rs.<amount> done using Credit Card". In the subclass UPIPayment, override the method to print "Payment of Rs.<amount> done using UPI". Accept user choice and amount, and display the payment result using method overriding. 4) Create a base class named Employee that contains a method displayDetails() which prints general employee details such as name and department. Now create a subclass Manager that inherits from Employee and overrides the displayDetails() method to include additional information such as the manager's team size or project name. In the main method, create objects of both Employee and Manager classes and call the displayDetails() method using each object to show how Java determines which version of the method to execute at runtime.	2	Ap

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5	**Program List 5:** 1) Describe abstract class called Shape which has three subclasses say Triangle, Rectangle and Circle. Define one method area() in the abstract class and override this area() in these three subclasses to calculate for specific object. 2) Write a program to create interface and declare one method interest(p,r,n) then create two class SimpleInterest and Compound Interest and calculate simple and compound interest. 3) Write a program to create package and import package.	2	Ap

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6	**Program List 6:** 1) Take the value of denominator and numerator from user using command-line argument. Implement the concept of exception handling to manage all possible run-time error. 2) Write a program to demonstrate throw and throws keyword.	3	Ap
7	**Program List 7:** 1) Design PrettyPrinter in a nonprimitive manner using Thread. 2) Write a Java program that creates two threads: ● First thread prints numbers from 1 to 10 at the interval of 1 second. ● Second thread prints numbers from 11 to 20 at the interval of 500 ms. Run both threads and display the output.	3	Ap
8	**Program List 8:** 1) Write a program to export book data in csv format. 2) Implement file read, write and append operation.	3	Ap
9	**Program List 9:** 1) Create ArrayList mover. In which selected element of one list moved to another. i.e. Subject Selected Subject Java DAA Maths-I EME Database Lets move Java and DAA to selected subject Subject Selected Subject	5	Ap

	Maths-I Java EME DAA Database 2) Create a generic class Box<T> with a method addItem (T item) that stores items in an ArrayList<T>. Demonstrate the class by: ● Creating a Box<String> for names. ● Creating a Box<Integer> for roll numbers. ● Display stored items for both.		
10	1) Write a javafx form for user signup which includes username(Textfield), password(passwordfield), gender(radiobutton),branch(combobox),address(textarea),hobbies(checkbox). Create two buttons display value of user data into a label. signup(button), clear(button). 2) Write a javafx program to move a circle from left to right when navigation keys been clicked. when mouse hovered over the circle then it change the color of circle.	**4**	**AP**

Faculty **HOD**